

EVENT NAME

<epic>

HACKATHON



CAMPUS PULSE

SMART CAMPUS. STUDENT WELL-BEING.

TEAM NAME : TEAM NOVA

PROJECT NAME : CAMPUS PULSE

****ONE APP. SMARTER MESS LINES, CALMER MINDS.****

TEAM MEMBERS:

NAME-dular solanki

Position-1st

Year-3rd

Branch-ECC

NAME- Vatsal

Position-2nd

Year-3rd

Branch-ECC

NAME-Devvrat Mishra

Position-3rd

Year-3rd

Branch-CE

**NAME-Chandrika
Solanki**

Position-4th

Year-3rd

Branch-IT

NAME-Isha Agarwal

Position-5th

Year-3rd

Branch-ECC

PROBLEM STATEMENT

PROBLEM STATEMENT ADDRESSED

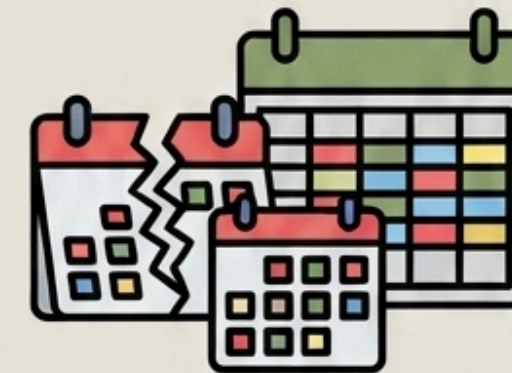
- **Mess food waste** —Campus messes cook food for a fixed number of students, even though daily attendance varies, leading to food wastag
- **Chaotic hostel allocation** —Room allocation ignores student preferences, department, and compatibility.Frequent room-change requests reduce student satisfaction
- **Resource scheduling** — Students can easily acces the schedule to check the sparity of the room and can have direct access to them for self study , no need to roome around .
- **Student Burnout** — During examination periods, students often struggle to balance classes, assignments, and exam preparation, leading to increased stress and academic burnout.
- **Most affected** — Hostel residents and final-year / exam-season students, who face all four pressures at once.



Mess food waste: Campus messes cook food for a fixed number of students, leading to food wastage.



Chaotic hostel allocation: Room allocation ignores student preferences, department, and compatibility.



Resource scheduling conflicts: Students cannot easily check or reserve study rooms and equipment.



Student Burnout: Increased stress and academic burnout due to examination pressures.

EXISTING SOLUTION



1. Smart Mess Management Systems (e.g., MealPe, campus QR meal-booking tools)

- ✔ **What it does:**
Track meal check-ins via QR/RFID to cut food waste.
- ✘ **What it misses:**
Only solves the mess food related problems
Food wastage is not monitored . As they are not capable for prior attendance of
The number of students having food.



2. Hostel Booking Portals (generic lab/room-booking ERP modules))

- ✔ **What it does:**
Let staff reserve rooms or equipment centrally.
- ✘ **What it misses:**
as it is done manually so there is less chance
To be done properly . Staff is also not able to
Check the availability as the remaining students
have wait for days .



3 AI guidance Software (e.g., SpaceBasic)

- ✔ **What it does**
guide the newcomers and are helpful to give the
- ✘ **What it misses:**
Most of the AI chatbots miss the deep briefing of
The newcomers and are not capable to guide properly.



4. Wellness & mental Health Platforms (e.g., Calm, Headspace, student wellness portals)

- ✔ **What it does:**
Help individuals plan, unwind, and manage stress.
- ✘ **What it misses:**
Not integrated with actual campus data
(Mess timings, Exam schedules, Room status)
Cannot proactively identify burnout risk based
on actual student workload.

YOUR INNOVATION

Campus Pulse AI – One Intelligent Platform for Smart Campus Management



PROPOSED SOLUTION

Campus Pulse AI unifies mess management, hostel allocation, resource scheduling, and student well-being into one AI-powered platform. It connects students, teachers, and administrators to enable smarter campus operations through real-time insights and intelligent recommendations.

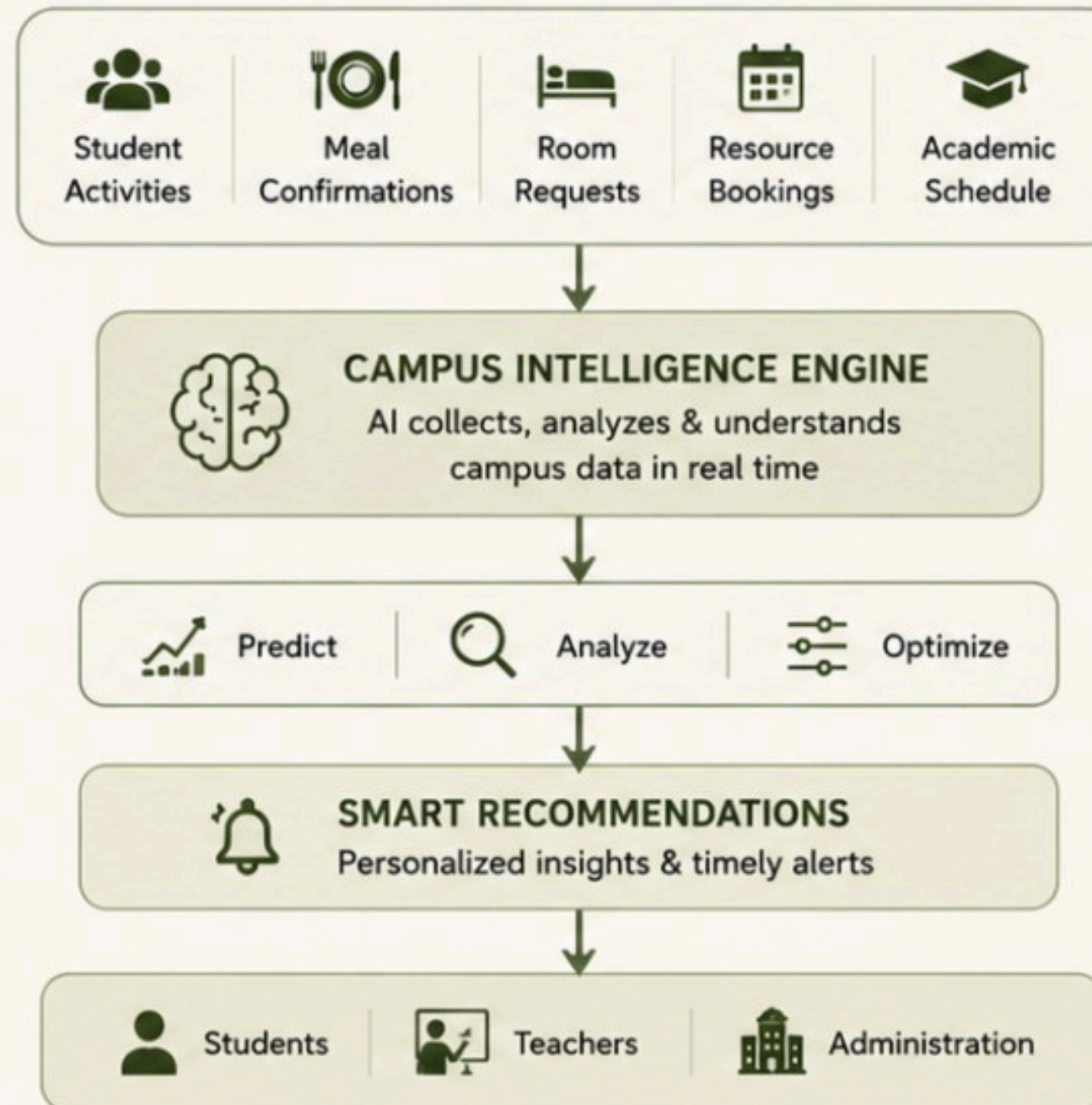


TARGET USERS

- Students** – Hostel residents & exam-season students
- Teachers** – Manage schedules, study materials & announcements
- Administrators** – Monitor hostel, mess & campus resources



HOW IT WORKS



EXPECTED IMPACT



20–30% reduction in food waste



30–40% fewer hostel room-change requests



Better utilization of campus resources



Improved student well-being & academic planning



Faster communication across campus



WHAT MAKES IT DIFFERENT

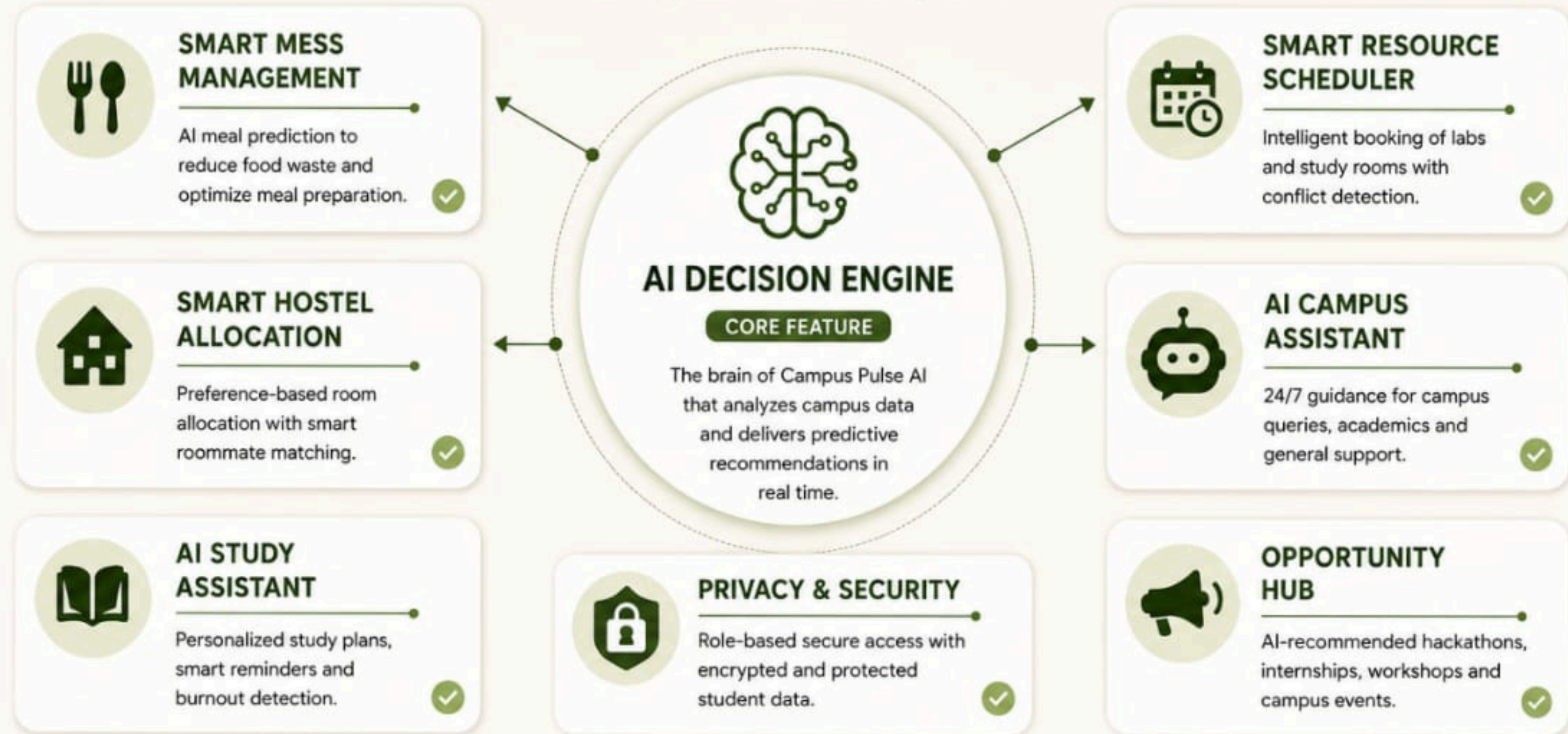
Unlike existing solutions that solve only one problem at a time, Campus Pulse AI integrates all major campus services into one intelligent ecosystem, enabling predictive decision-making, real-time collaboration, and personalized recommendations for the entire campus.



Campus Pulse AI transforms disconnected campus services into one AI-powered ecosystem, enabling smarter decisions, efficient resource utilization, and a better campus experience for every student.

FEATURES

Powering a Smarter Campus with AI



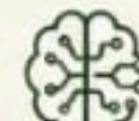
KEY BENEFITS



ONE PLATFORM
All campus services in one place.



REAL-TIME NOTIFICATIONS
Instant alerts and important updates.



AI-POWERED INSIGHTS
Data-driven decisions for better outcomes.



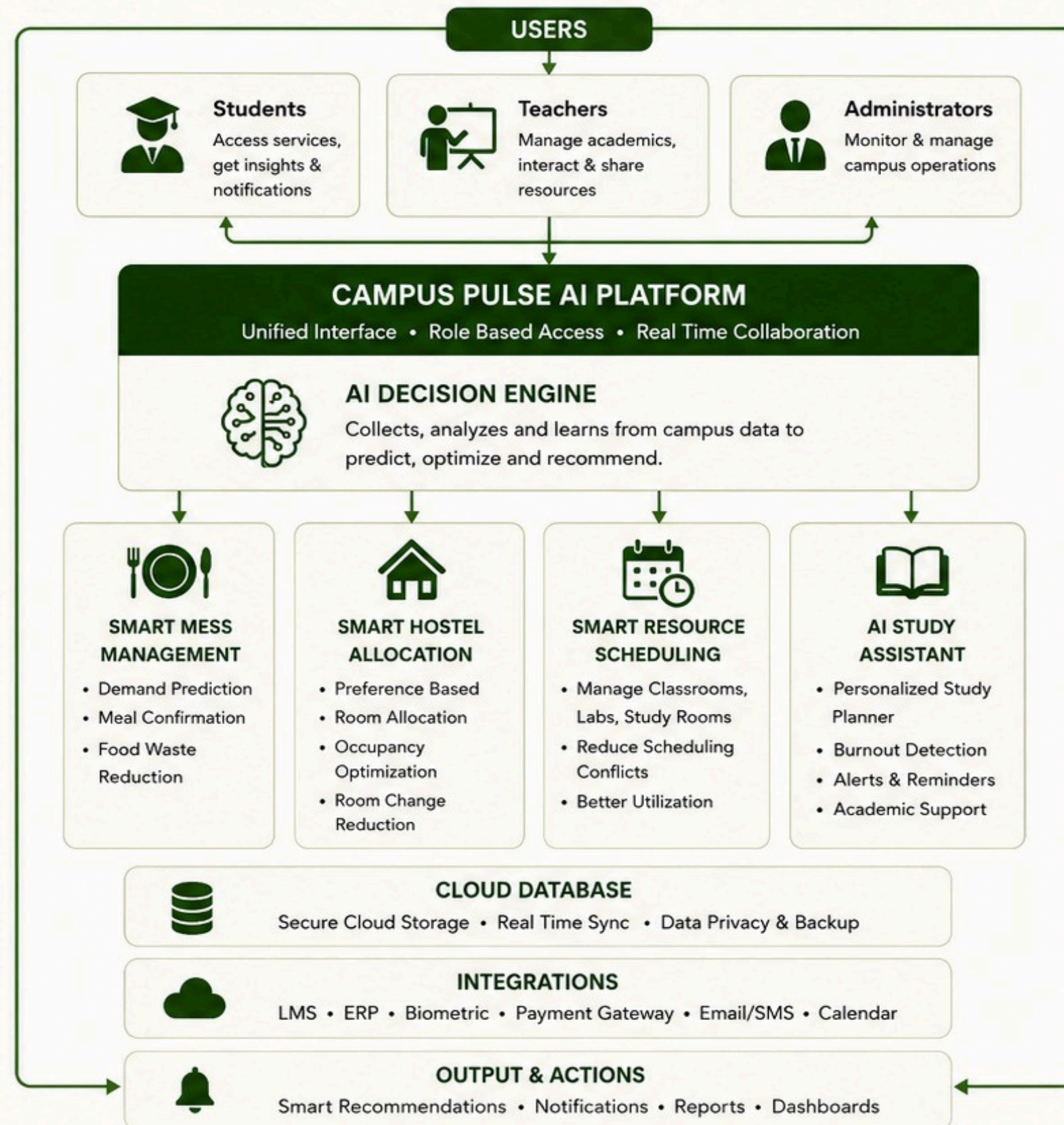
BETTER RESOURCE UTILIZATION
Optimize facilities and reduce waste.



IMPROVED STUDENT EXPERIENCE
Smarter services for a better campus life.

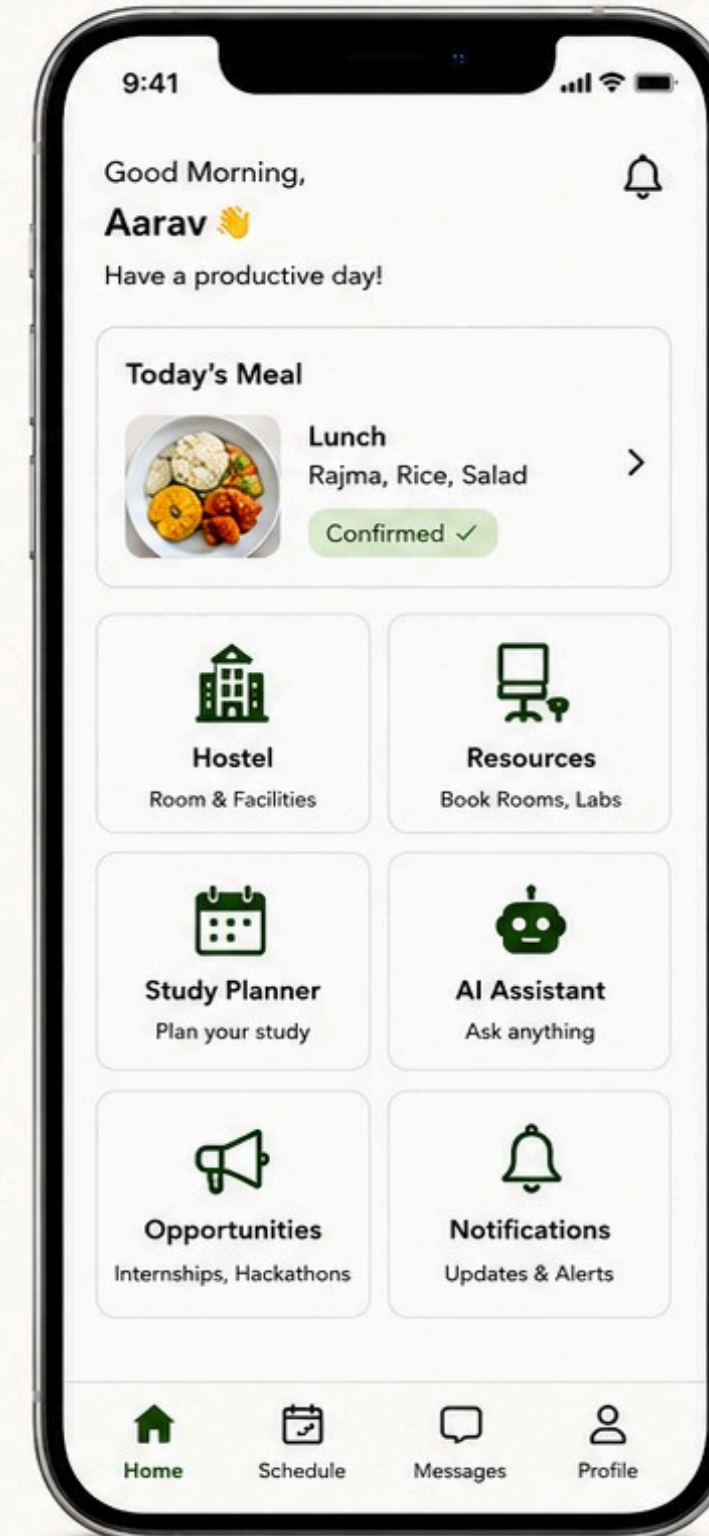
SYSTEM ARCHITECTURE

Campus Pulse AI – One Intelligent Platform for Smart Campus Management



PROTOTYPE (WIREFRAME)

Student Dashboard (Mobile View)



HOW USERS INTERACT

- Students access all services, get AI recommendations & real-time updates.
- Teachers manage academics, upload materials, and communicate.
- Administrators get insights, analytics and manage campus operations.

KEY HIGHLIGHTS

- ✓ Role based access
- ✓ AI powered insights
- ✓ Real time collaboration
- ✓ Scalable & Secure
- ✓ All in one platform

SOLUTION COMPARISON

CampusAI Companion vs Existing Campus Apps

FEATURE	EXISTING APPS	CAMPUSAI COMPANION
Single unified platform (mess + hostel + wellbeing + resources)	✗ No — needs 4-5 separate platforms	✓ Yes — one single app
Data privacy & secure access	⚠ Varies by app, no unified control	✓ Centralized, secure login
AI chatbot guidance, 24x7	✗ Not properly applied	✓ Built-in AI chatbot
Peer & teacher chat section	⚠ Fragmented, app-specific only	✓ Integrated chat, all in one place
Centralized news portal for students	✗ Not available or never works properly	✓ Real-time alerts, before they're "old news"
Digital library access	⚠ Separate ERP portal only	✓ Built-in library access

No more missed opportunities — get hackathon & competition alerts before anyone else does.

FUTURE IMPROVEMENTS

A SEPERATE APP THAT IS DIRECTLY CONNECTED WITH ALL THE RESOURCES SEPARATLY AND CAN DEIRECTLY ACCESS DIRECTLY FROM YOUR PHONE.

FOLLOWING WILL BE THE FUTURE FEATURES :

- **ATTENDANCES**
- **DOUBT SESSIONS**
- **FEES SUBMISSIONS**

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THANK YOU

TEAM NOVA — CAMPUS PULSE AI